SAM-FuseNet: Segment Anything Guided Multimodal Fusion for RGB–Thermal Aerial Robotic Perception

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***Abstract*—Multimodal semantic segmentation with RGB and Thermal (RGB–T) inputs is critical for robust aerial robotic perception, yet existing methods face challenges in effectively adapting pretrained representations and balancing complementary cues under variable illumination. To overcome these challenges, we introduce segment anything guided multimodal fusion network (SAM-FuseNet), a novel RGB—T framework that leverages modality-specific encoders built on the segment anything model 2 (SAM2) vision transformer (ViT) backbone, enhanced with low-rank adaptation (LoRA) modules for efficient modality-specific fine-tuning without full model retraining. Cross-modal fusion is achieved via a pixel-wise spatial attention mechanism combined with channel recalibration, allowing adaptive weighting of contributions from each modality. A dual-pathway segmentation head further improves prediction quality: the first pathway progressively refines coarse-to-fine masks under SAM2 guidance, while the second pathway performs hierarchical feature fusion to capture long-range contextual information. Extensive experiments on the Caltech RGB–T and PST900 benchmarks demonstrate the effectiveness of SAM-FuseNet, achieving improvements of up to 3.93% in mean intersection-over-union (mIoU) while reducing the trained model parameters by more than 50%. Qualitative results further demonstrate sharper boundaries, robustness under underexposed conditions, and resilience to modality degradation.**

***Index Terms*—Aerial robotics, multimodal semantic segmentation, RGB–thermal (RGB–T), segment anything model (SAM).**

NOMENCLATURE

Abbreviation	Expansion
$\mu \in \{\text{RGB, Thermal}\}$	Modality index.
$\ell \in \{c, m, f\}$	Scale index: coarse, mid, fine.
$k \in \{1, \ldots, \mathcal{C}\}$	Semantic class index.
$\mathbf{X}^{(\mu)} \in \mathbb{R}^{N \times d}$	Token features for modality μ.
$F_\ell^{(\mu)} \in \mathbb{R}^{c_\ell \times H_\ell \times W_\ell}$	Encoder feature at scale ℓ.
$F_\ell^{(\text{fused})'}$	SE-recalibrated fused feature map.
Q, K, V	Query, key, value projections.
W_q, W_k, W_v	Projection matrices ($d \times d_k, d \times d_k, d \times d_v$
$\Delta W_q, \Delta W_v$	LoRA low-rank updates for Q, V.
$\text{Up}^{(H,W)}$	Bilinear upsampling to (H, W).
[U|V]	Channel-wise concatenation.
BN, δ, σ	BatchNorm, ReLU, sigmoid.
$\text{Conv}_{a\times b}^{c}$	Conv with kernel $a \times b, c$ outputs.
MLP_ℓ	Scale-specific MLP to d_{dec} dims.
$\arg\max_k$	Per-pixel class prediction over k.
$\mathcal{L}_{\text{CE}}, \mathcal{L}_{\text{Ohem}}$	Cross-entropy loss; OHEM variant.

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I. INTRODUCTION

SEMANTIC segmentation with paired RGB and thermal (RGB–T) sensing is a key capability for aerial robotic platforms operating in natural environments where illumination changes, shadows, and low-texture regions often reduce the reliability of RGB-only perception. Thermal imaging provides complementary cues when color and texture are weak by highlighting temperature-related differences that support the separation of materials or objects with similar RGB appearance [1], [2], [3]. Despite this complementarity, reliable RGB–T segmentation for aerial systems remains difficult. Aerial viewpoints induce large scale variation and small object footprints; platform motion introduces blur and rolling-shutter artifacts; and RGB–T cameras typically differ in optics, field-of-view, and resolution, which leads to cross-modal misalignment that varies across the image. Thermal measurements also depend on time and weather, which alters their statistical properties and undermines fixed fusion rules.

Current RGB–T segmentation approaches generally use explicit cross-modal fusion with modality-specific encoders followed by feature combination [1], [2], [3]. RTFNet, for example, employs parallel encoders and feature summation to increase robustness under poor illumination [1]. Later designs such as CAINet [4] and CMX [5] incorporate global reasoning to strengthen cross-modal interaction. However, many pipelines still rely on full fine-tuning of large backbones such as DeepLabv3+ [6], PSPNet [7], or transformer-based models including SegFormer [8]. This increases training and deployment cost and creates tension with resource constraints of aerial platforms. Furthermore, fusion is often implemented

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with spatially uniform or weakly adaptive schemes that do not adequately react to spatially varying modality reliability, misalignment, or thermally induced appearance changes. These limitations lead to reduced precision for boundaries and small structures.

Vision foundation models such as the Segment Anything Model (SAM) [9] and SAM2 provide strong pretrained representations, yet directly adapting them for RGB–T semantic segmentation raises two challenges. First, full fine-tuning of SAM-scale backbones is expensive. Second, single-modal priors do not align with the statistical properties of multimodal inputs. Parameter-efficient fine-tuning (PEFT) methods such as low-rank adaptation (LoRA) [10] and SAM-Adapter [11] address the first issue by updating small low-rank modules instead of all model parameters. For RGB–T segmentation, OpenRSS [12] introduces LoRAs to improve efficiency, and MLE-SAM [13] extends SAM2 to heterogeneous modalities with a Mixture-of-LoRA-Experts that performs global modality-level routing. Although effective for reducing adaptation cost, these methods focus on global aggregation and do not explicitly model pixel-wise or scale-wise fusion, which is essential for dense RGB–T aerial scenes. Spatial reliability of each modality varies across the field of view and at different feature scales, and ignoring these variations limits spatial precision and boundary accuracy.

To address these limitations, we propose the segment anything guided multimodal fusion network (SAM-FuseNet) for RGB–T semantic segmentation in aerial robotic perception. The framework contains three stages. The first stage applies modality-specific LoRA modules to a SAM2 ViT backbone to adapt pretrained representations to RGB–T data while maintaining low trainable parameter count. The second stage introduces pixel-wise spatial attention fusion at multiple feature scales, combined with channel-level recalibration to suppress unreliable modality evidence and strengthen locally consistent cues. The third stage uses a dual-pathway segmentation head. One pathway extends the SAM2 mask-decoder with coarse-to-fine logit refinement guided by multiscale fused features. The other pathway aggregates contextual information through hierarchical fusion and dilated convolutions to improve long-range semantics. Unlike global routing strategies such as MLE-SAM, SAM-FuseNet performs spatially varying modality weighting at every scale, aligning the fusion process with the heterogeneous reliability patterns present in aerial RGB–T imagery. Multiscale fusion at coarse, mid, and fine resolutions supports joint modeling of global semantics, region layout, and detailed boundaries. A fine-scale-only variant is additionally evaluated to quantify the benefits of incorporating coarser scales.

The main contributions of this work are as follows.

1) A SAM2-based RGB–T segmentation framework that integrates modality-specific LoRA modules into transformer self-attention, enabling efficient adaptation of a foundation backbone to aerial RGB–Thermal data without full fine-tuning.
2) A pixel-wise and scale-wise fusion mechanism that combines spatial attention with squeeze-and-excitation (SE) recalibration to achieve localized RGB–T integration under cross-modal misalignment and thermally induced appearance variation.
3) A dual-pathway segmentation head that extends the SAM2 mask-decoder with coarse-to-fine refinement using fused multiscale features and complements it with hierarchical fusion and dilated contextual aggregation for robust semantic reasoning.
4) Extensive experiments and controlled ablations that validate the contribution of multiscale fusion and show consistent improvements over strong RGB–T baselines.

II. Related Work

A. Classical RGB–T Semantic Segmentation Networks

Early RGB–T semantic segmentation models typically processed RGB–T modalities with independent encoders and fused features at later stages. RTFNet adopted parallel ResNet encoders with element-wise summation, yielding strong performance under poor illumination [1]. ABMDRNet explicitly minimized modality discrepancies through adaptive-weighted bi-directional bridges, integrating multiscale context modules and surpassing MFNet [3]. LASNet leveraged complementary cues via collaborative location, activation, and sharpening modules with auxiliary supervision to improve convergence [2]. Label-Guided Real-Time Fusion employed semantic-label-based spatial attention for efficient integration of multilevel features, striking a balance between accuracy and computational efficiency [14].

Beyond RGB–T segmentation, multimodal remote sensing research provides complementary evidence regarding cross-domain and cross-sensor perception. SpectralGPT proposed a spectral remote sensing foundation model and showed that large-scale pretraining can support generalization across diverse sensing conditions [15]. The Cross-City benchmark [16] advanced multimodal semantic segmentation under geographic and environmental shifts, demonstrating the difficulty of transferring models across regions. Earlier analyses of multimodal convolutional networks for remote sensing classification [17] further indicated that integrating heterogeneous sources strengthens representation quality while introducing alignment and fusion challenges similar to those in RGB–T segmentation. These findings underscore the need for multimodal frameworks that maintain reliability across domains and sensing conditions.

Subsequent methods emphasized explicit cross-modal interaction. CAINet introduced complementary reasoning, global context aggregation, and detail refinement, achieving state-of-the-art performance on MFNet and PST900 [4]. CMX extended robustness to multiple modalities, introducing cross-modal rectification and dedicated fusion modules for RGB, Thermal, depth, and LiDAR [5]. SGFNet incorporated spectral-aware fusion to capture multifrequency signals, improving accuracy [18]. Complementary random masking (CRM) further mitigated modality reliance by randomly masking input modalities during training with self-distillation [19].

Robustness to missing or degraded modalities has also become a central theme. CMNeXt introduced the DeLiVER dataset with RGB, LiDAR, event, multiview, and Thermal

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sensors, and proposed a Self-Query Hub for adaptive multimodal fusion [20]. Broader benchmark studies [21] reinforced the importance of architectures resilient to sensor failures and highlighted the need for models capable of maintaining high performance despite noisy or incomplete modalities.

Many RGB–T models are built upon backbones originally developed for RGB segmentation. Fully convolutional networks (FCNs) demonstrated dense prediction from classification networks by employing skip connections to merge semantic and spatial details [22]. DeepLabv3+ introduced atrous separable convolutions with decoders for improved boundary delineation and multi-scale context [6]. PSPNet aggregated global context via pyramid pooling [7], while SegFormer combined hierarchical transformers with lightweight decoders, achieving high accuracy with fewer parameters [8]. These advances laid the foundation for RGB–T segmentation, but adapting them to multimodal data remains challenging.

B. SAM for Semantic Segmentation

The SAM leverages vision transformers (ViTs) for prompt-driven zero-shot segmentation [9]. However, SAM struggles with modality discrepancies and cannot directly handle multimodal fusion. To adapt SAM for RGB–T segmentation, recent work has proposed encoder refinements and prompting mechanisms.

OpenRSS extended SAM with thermal information prompt (TIP) modules and dynamic LoRA (DLoRA), integrating SAM and CLIP for open-vocabulary segmentation [12]. Prompt-based designs such as Mask2Former employed masked attention for task unification [23], while SAM-Adapter introduced lightweight adapters and prompts directly into SAM's encoder [11]. TokenFusion dynamically substituted and pruned token-level representations, enhancing multimodal alignment while preserving spatial structure [24]. These works highlight the potential of adapting SAM to multimodal settings, though challenges remain in efficiently bridging modality gaps.

Within remote sensing, MeSAM modifies both encoder and decoder to preserve high-frequency details and to add multiscale links tailored to optical scenes, reporting consistent gains over SAM baselines [25]. PointSAM targets annotation cost by fine-tuning SAM with point supervision using self-training with prototype matching and negative prompt calibration [26]. Qiao et al. [27] propose a training-free enhancement that uses SAM-derived object and boundary cues to refine predictions without retraining the base segmentation model or the foundation model, improving boundary quality on remote sensing benchmarks. For change detection under limited labels, Gao et al. [28] integrate a small adapter with a SAM-family backbone to adapt with few samples, reducing data demands while retaining mask quality. For hyperspectral multiclass change detection, Zhan et al. [29] couple a spatial–spectral Mamba module with SAM-guided priors to stabilize category transitions across time and spectrum.

C. PEFT Techniques for SAM

The high computational cost of fully fine-tuning SAM motivates PEFT. LoRA inserts low-rank trainable matrices into pretrained weights, reducing parameter overhead without sacrificing accuracy [10]. Extensions of LoRA demonstrate strong adaptability in multimodal tasks: CWSAM employed class-specific adapters and mask decoders for synthetic aperture radar imagery [30], while MLE-SAM integrated modality-specific LoRA experts, maintaining robustness even with partially missing inputs on benchmarks like DeLiVER and MUSES [13]. OpenRSS also applied DLoRA to enhance RGB–T adaptation [12].

Prompt and adapter-based methods further expand PEFT. Visual prompt tuning (VPT) learned prompts with a frozen backbone, outperforming full fine-tuning [31]. Distribution-Aware VPT adapted prompts with metric-guided adjustment [32]. SAM-Adapter combined prompts with adapters inside SAM's encoder [11], while VT-Adapter [33], META [34], and sensitivity-aware PEFT (SPT) [35] validated the broader effectiveness of adapter-based strategies. Together, these methods highlight the efficiency and flexibility of PEFT for large multimodal foundation models.

Despite steady progress, several challenges remain open in RGB–T semantic segmentation. First, classical fusion networks often rely on late feature aggregation, which limits the ability to fully exploit cross-modal interactions. Second, transformer-based SAM adaptations struggle with modality discrepancies, as SAM was originally designed for unimodal RGB data. Third, existing PEFT techniques demonstrate efficiency, but they are not always optimized for fine-grained multimodal fusion, particularly under aerial robotics constraints where robustness to illumination variation, sensor noise, and computational efficiency are critical.

The proposed SAM-FuseNet framework addresses these challenges by introducing LoRA-enhanced modality-specific encoders built on the SAM2 Vit backbone. We adapt pretrained representations to both RGB–T modalities without requiring full fine-tuning. In contrast to prior PEFT approaches that remain task-agnostic, SAM-FuseNet explicitly tailors LoRA modules and fusion mechanisms to RGB–T segmentation, thereby combining parameter efficiency with strong multimodal robustness. This design enables consistent performance despite noisy or missing modalities, making it well-suited for aerial robotic applications in natural environments.

III. Preliminary

The SAM2 is a transformer-based instance segmentation framework [36] that consists of three major components: a hierarchical image encoder, a feature pyramid network (FPN) neck, and a mask decoder. The encoder adopts the Hiera design [37] to generate multi-scale feature representations. Specifically, the input image is first tokenized into patch embeddings and then processed through successive stages, where each stage reduces spatial resolution by half while doubling the channel dimension. Within each stage, windowed multihead self-attention (MHSA) layers and pooling operations are employed to capture both spatial and semantic dependencies across scales. The FPN aligns channel dimensions and aggregates features from different depths via lateral connections and a top-down pathway, yielding a unified multiscale representation. To retain spatial awareness, sinusoidal positional

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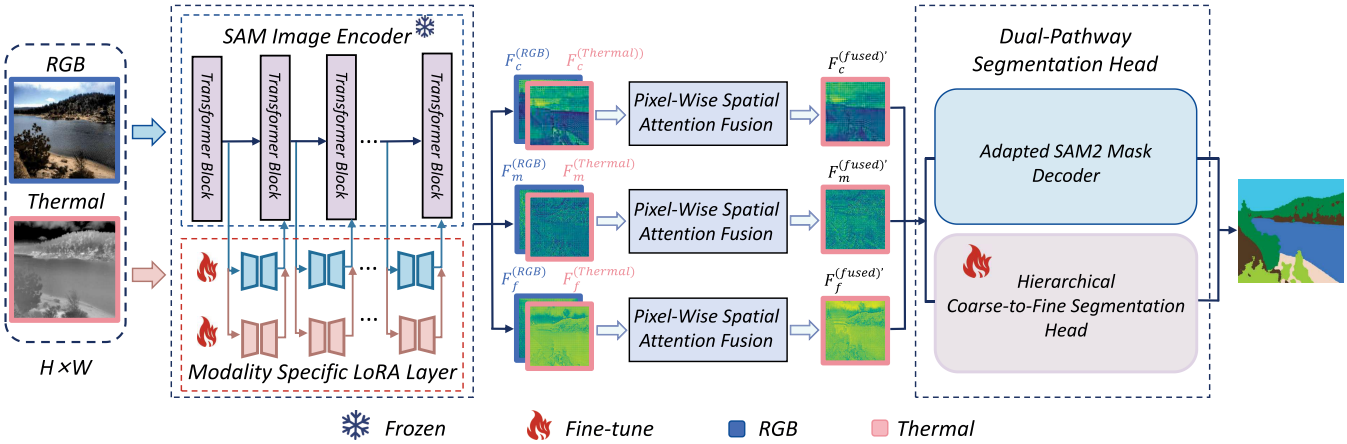


Fig. 1. Framework overview of SAM-FuseNet for RGB–T semantic segmentation. The model integrates LoRA-enhanced modality-specific encoders, a spatial attention fusion module with SE blocks, and a dual-pathway segmentation head.

encodings are incorporated into the fused features. The mask decoder is built on transformer cross-attention, where a set of learnable mask tokens interacts with the fused features and positional encodings. These mask tokens are iteratively refined by stacked cross-attention and feed-forward layers. Finally, an upsampling head restores the spatial resolution, producing high-quality segmentation masks. Multiple masks are predicted to effectively handle overlapping objects and complex scenes.

IV. Framework Description

A. SAM-FuseNet Framework Overview

We propose the SAM-FuseNet framework for multimodal semantic segmentation using RGB–T images, with a focus on aerial robotic perception in natural environments. The core of our framework is a modality-specific encoder constructed upon the SAM2 ViT backbone, enhanced with LoRA modules. By inserting LoRA into the self-attention layers, we adapt pretrained representations for each modality without the computational overhead of full model fine-tuning. Each encoder extracts hierarchical features at coarse, intermediate, and fine spatial scales, which are then integrated by a pixel-wise spatial attention module. This module learns attention maps to dynamically weight the contribution of each modality, while SE blocks recalibrate channel responses to strengthen complementary information. The fused multiscale features are subsequently processed by a dual-pathway hierarchical segmentation head. The first pathway extends the SAM2 mask decoder by refining coarse predictions in a progressive manner: logits derived from the coarse fused feature are sequentially updated using mid- and fine-level fused features, producing high-resolution segmentation masks through iterative refinement. The second pathway complements this process with an explicit coarse-to-fine fusion strategy. Fused features at each scale are projected into a latent space using scale-specific multi-layer perceptron (MLP), hierarchically integrated through convolutional fusion to capture long-range contextual semantics. Both pathways generate segmentation predictions at full resolution, which together improve robustness and fine-grained accuracy. As depicted in Fig. 1, the overall architecture illustrates the interaction between key components, highlighting how RGB–T modalities jointly enhance robustness and semantic segmentation performance. The main notations are summarized in Nomenclature.

B. LoRA-Enhanced Modality-Specific Encoder

To effectively integrate complementary information from RGB–T modalities, we propose a modality-specific encoding framework constructed on top of the SAM2 ViT backbone. To leverage the pretrained semantic representations of SAM2 without incurring the computational cost of full fine-tuning, we incorporate LoRA modules into the MHSA layers of the transformer. Specifically, each modality $\mu \in \{\text{RGB}, \text{Thermal}\}$ is processed by an independent encoder $E^{(\mu)}$, which converts the raw modality input $I^{(\mu)}$ into a sequence of patch embeddings. These embeddings are subsequently augmented with positional encodings, producing tokenized features $\mathbf{X}^{(\mu)} \in \mathbb{R}^{N\times d}$, where N represents the number of tokens and d denotes the embedding dimension.

Within the transformer attention mechanism, the tokenized features $\mathbf{X}^{(\mu)}$ are used to generate queries (Q), keys (K), and values (V) through linear projections shown in the following equation:

$$Q = \mathbf{X}^{(\mu)} W_q, \quad K = \mathbf{X}^{(\mu)} W_k, \quad V = \mathbf{X}^{(\mu)} W_v \tag{1}$$

where $W_q, W_k \in \mathbb{R}^{d\times d_k}$ and $W_v \in \mathbb{R}^{d\times d_v}$. Thus $Q, K \in \mathbb{R}^{N\times d_k}$ and $V \in \mathbb{R}^{N\times d_v}$. Standard scaled dot-product attention is then formulated as the following equation:

$$\text{Attention}\,(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V \tag{2}$$

where d_k is the dimension of the key vectors.

To enable modality-specific adaptation without full fine-tuning, LoRA introduces trainable low-rank matrices applied to the query and value projections, while the key projection remains fixed. Let $W_{A,q}, W_{A,v} \in \mathbb{R}^{d\times r}, W_{B,q} \in \mathbb{R}^{r\times d_k}$,

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$W_{B,v} \in \mathbb{R}^{r\times d_v}, r \ll d$. The low-rank updates are defined as the following equation:

$$\Delta W_q = W_{A,q}W_{B,q}, \qquad \Delta W_v = W_{A,v}W_{B,v}. \tag{3}$$

Here, $W_{A,q}$ and $W_{A,v}$ are randomly initialized, while $W_{B,q}$ and $W_{B,v}$ are initialized to zeros to preserve the functionality of the pretrained SAM2 model at initialization. The adapted projections are then expressed as the following equation:

$$Q = \mathbf{X}^{(\mu)}\left(W_q + \Delta W_q\right), \qquad V = \mathbf{X}^{(\mu)}\left(W_v + \Delta W_v\right). \tag{4}$$

Following the design of SAM2, the encoder first applies a patch embedding to project $I^{(\mu)}$ into an initial latent map, and then processes it through hierarchical stages with progressive downsampling and channel expansion. Each stage employs local self-attention within shifted windows to refine features, and the outputs of selected stages are aggregated by the FPN to form multiscale features $\{F_c^{(\mu)}, F_m^{(\mu)}, F_f^{(\mu)}\}$, where $F_c^{(\mu)} \in \mathbb{R}^{d\times H_c\times W_c}$ corresponds to coarse-level features with low spatial resolution but high semantic abstraction. The intermediate and fine-grained features, $F_m^{(\mu)}$ and $F_f^{(\mu)}$, are extracted from subsequent layers of the FPN, capturing progressively finer semantic details. We use (H_ℓ, W_ℓ) to denote the spatial resolution at scale $\ell \in \{c, m, f\}$, where $H_m = 2H_c$, $W_m = 2W_c$, $H_f = 4H_c$, and $W_f = 4W_c$. The original input has size (H_0, W_0). Concretely, coarse-level features typically have dimensions $d \times H_c \times W_c$ (e.g., $256 \times 64 \times 64$), mid-level features $(d/4) \times H_m \times W_m$ (e.g., $64 \times 128 \times 128$), and fine-level features $(d/8) \times H_f \times W_f$ (e.g., $32 \times 256 \times 256$).

These modality-specific multiscale representations are subsequently fed into a dedicated spatial fusion module designed to align and integrate the information across RGB–T modalities. The fusion module constructs multimodal feature representations that enhance semantic segmentation performance by combining complementary visual and thermal cues.

C. Pixel-Wise Spatial Attention Fusion

Formally, for each modality $\mu \in \{\text{RGB}, \text{Thermal}\}$, the encoder produces multi-scale features $F_\ell^{(\mu)} \in \mathbb{R}^{c_\ell\times H_\ell\times W_\ell}$. At spatial position (i, j), we extract the feature vector $\mathbf{f}_{ij}^{(\mu,\ell)} = [F_\ell^{(\mu)}]_{,i,j} \in \mathbb{R}^{c_\ell}$ and normalize it using layer normalization to obtain $\tilde{\mathbf{f}}_{ij}^{(\mu,\ell)}$. The normalized vectors are concatenated to form,as demonstrated in the following equation:

$$\mathbf{z}_{ij}^{(\ell)} = \left[\tilde{\mathbf{f}}_{ij}^{(\text{RGB},\ell)} \,\|\, \tilde{\mathbf{f}}_{ij}^{(\text{Thermal},\ell)}\right] \in \mathbb{R}^{2c_\ell}. \tag{5}$$

A lightweight spatial attention network $g^{(\ell)}$, implemented as a two-layer MLP with 1×1 convolutions, processes $\mathbf{z}_{ij}^{(\ell)}$ to output two modality-specific logits $\ell_{ij,\text{RGB}}^{(\ell)}$ and $\ell_{ij,\text{Thermal}}^{(\ell)}$. The first layer reduces the channel dimension by a factor r_d and applies a ReLU activation, while the second restores dimensionality to 2 for the logits. These logits are normalized by a temperature-scaled softmax described in the following equation:

$$\alpha_{ij,\mu}^{(\ell)} = \frac{\exp\left(\ell_{ij,\mu}^{(\ell)}/\tau\right)}{\exp\left(\ell_{ij,\text{RGB}}^{(\ell)}/\tau\right) + \exp\left(\ell_{ij,\text{Thermal}}^{(\ell)}/\tau\right)}, \quad \sum_\mu \alpha_{ij,\mu}^{(\ell)} = 1 \tag{6}$$

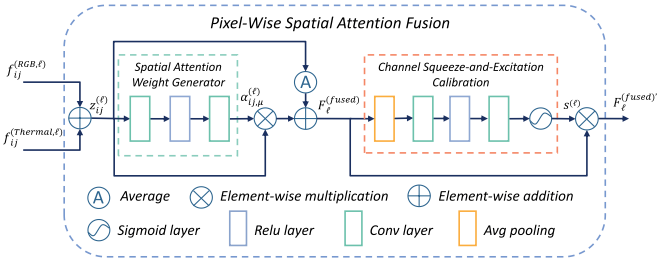


Fig. 2. Modality-aware fusion with spatial attention and channel recalibration for refined feature representation.

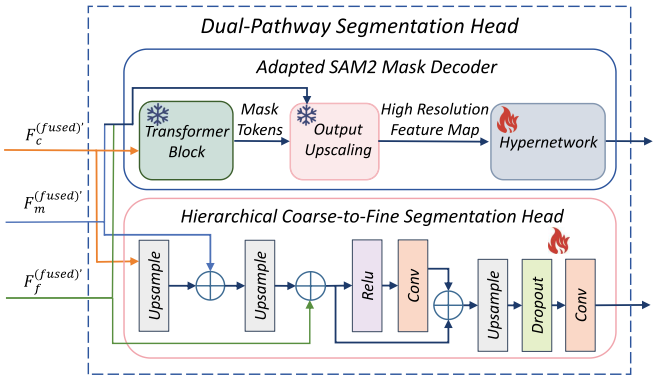


Fig. 3. Dual-pathway segmentation head combining SAM2-guided refinement and hierarchical fusion.

where $\tau > 0$ is a learnable temperature parameter controlling the sharpness of the distribution.

The fused feature vector at (i, j) is computed as the following equation:

$$\mathbf{f}_{ij}^{(\ell,\text{fused})} = \sum_{\mu\in\{\text{RGB},\text{Thermal}\}} \left(\frac{1-\lambda}{2} + \lambda\,\alpha_{ij,\mu}^{(\ell)}\right)\tilde{\mathbf{f}}_{ij}^{(\mu,\ell)}, \quad \lambda \in [0, 1] \tag{7}$$

which ensures a convex combination of modality contributions, avoiding uncontrolled scaling. Aggregating across all spatial positions reconstructs the fused feature map $F_\ell^{(\text{fused})} \in \mathbb{R}^{c_\ell\times H_\ell\times W_\ell}$.

To further refine feature quality, we employ a SE block. Global average pooling is applied over $F_\ell^{(\text{fused})}$ to compute a channel descriptor $u_k^{(\ell)} = 1/H_\ell W_\ell \sum_{i=1}^{H_\ell}\sum_{j=1}^{W_\ell}[F_\ell^{(\text{fused})}]_{k,i,j}$, which forms the vector $\mathbf{u}^{(\ell)} \in \mathbb{R}^{c_\ell}$. This descriptor is passed through two fully connected layers with reduction ratio r_c, followed by ReLU and sigmoid activations, to yield channel-wise recalibration weights, defined in the following equation:

$$\mathbf{s}^{(\ell)} = \sigma\left(W_2^{(\ell)}\,\delta\left(W_1^{(\ell)}\mathbf{u}^{(\ell)}\right)\right) \tag{8}$$

where $W_1^{(\ell)} \in \mathbb{R}^{(c_\ell/r_c)\times c_\ell}$, $W_2^{(\ell)} \in \mathbb{R}^{c_\ell\times(c_\ell/r_c)}$, $\delta(\cdot)$ is ReLU, and $\sigma(\cdot)$ is sigmoid. The recalibrated fused map is then formalized in (9). The network architecture is shown in Fig. 2.

$$F_\ell^{(\text{fused})'} = \mathbf{s}^{(\ell)} \odot F_\ell^{(\text{fused})}. \tag{9}$$

D. Dual-Pathway Design of Segmentation Head

As shown in Fig. 3, in the dual-pathway design, the first pathway extends the original SAM2 mask decoder by

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progressively refining multi-scale fused features. The coarse fused feature $F_c^{(\text{fused})'}$ is first processed by a transformer-based decoder f_{dec} to produce coarse logits $\mathbf{S}_c^0$. These logits are then sequentially refined using mid- and fine-level fused features, each projected to $\mathcal{C}$ semantic channels with a 1×1 convolution and bilinearly upsampled to the target resolution,which is mathematically proven in the following equation:

$$\begin{aligned}\mathbf{S}_c^0 &= f_{\text{dec}}\left(F_c^{(\text{fused})'}\right)\\ \mathbf{S}_m^0 &= \text{Up}^{(H_m,W_m)}\left(\mathbf{S}_c^0\right) + \text{Conv}_{1\times1}^{\mathcal{C}}\left(F_m^{(\text{fused})'}\right)\\ \mathbf{S}_f^0 &= \text{Up}^{(H_f,W_f)}\left(\mathbf{S}_m^0\right) + \text{Conv}_{1\times1}^{\mathcal{C}}\left(F_f^{(\text{fused})'}\right)\\ \bar{\mathbf{S}}_0 &= \text{Up}^{(H_0,W_0)}\left(\mathbf{S}_f^0\right).\end{aligned} \tag{10}$$

The final segmentation mask is obtained by taking the pixel-wise maximum across classes described in the following equation:

$$\hat{\mathbf{Y}}_0(i,j) = \arg\max_{k\in\{1,\ldots,\mathcal{C}\}} \bar{\mathbf{S}}_0^{(k)}(i,j) \quad \forall (i,j)\in[H_0]\times[W_0]. \tag{11}$$

The second pathway complements the first pathway, whose segmentation head employs explicit hierarchical feature fusion with contextual enhancement. Each fused feature map $F_\ell^{(\text{fused})'}$ is first projected into a latent space of dimension d_{dec} using a scale-specific MLP, producing $\mathbf{S}_\ell^1$ at each scale. Hierarchical fusion is then performed in a coarse-to-fine manner as demonstrated in the following equation:

$$\begin{aligned}\mathbf{S}_c^1 &= \text{MLP}_c\left(F_c^{(\text{fused})'}\right),\\ \mathbf{S}_m^1 &= \text{ReLU}\left(\text{BN}\left(\text{Conv}_{3\times3}^{d_{\text{dec}}}\left(\left[\ \text{Up}^{(H_m,W_m)}\left(\mathbf{S}_c^1\right)\right.\right.\right.\right.\\ &\qquad \left.\left.\left.\left.\Vert\ \text{MLP}_m\left(F_m^{(\text{fused})'}\right)\ \right]\right)\right)\right)\\ \mathbf{S}_f^1 &= \text{ReLU}\left(\text{BN}\left(\text{Conv}_{3\times3}^{d_{\text{dec}}}\left(\left[\ \text{Up}^{(H_f,W_f)}\left(\mathbf{S}_m^1\right)\right.\right.\right.\right.\\ &\qquad \left.\left.\left.\left.\Vert\ \text{MLP}_f\left(F_f^{(\text{fused})'}\right)\ \right]\right)\right)\right).\end{aligned} \tag{12}$$

To enhance long-range dependencies and expand the effective receptive field without increasing spatial resolution, multiple parallel dilated convolutions with varying rates are applied to $\mathbf{S}_f^1$, and their outputs are concatenated and linearly projected to form class logits $\mathbf{S}^1$. After upsampling to (H_0, W_0), the segmentation mask is then derived by pixel-wise maximization as shown in the following equation:

$$\bar{\mathbf{S}}_1 = \text{Up}^{(H_0,W_0)}\left(\mathbf{S}^1\right), \quad \hat{\mathbf{Y}}_1(i,j) = \arg\max_{k\in\{1,\ldots,\mathcal{C}\}} \bar{\mathbf{S}}_1^{(k)}(i,j). \tag{13}$$

The ground-truth segmentation labels $\mathbf{L}\in\mathbb{R}^{H\times W}$ are defined such that $\mathbf{L}(i,j)\in\{0,1,\ldots,\mathcal{C}-1,255\}$, where 255 indicates the ignore label. The OhemCrossEntropy loss for a single prediction map $\mathbf{S}$ is given as the following equation:

$$\mathcal{L}_{\text{Ohem}}(\mathbf{S},\mathbf{L}) = \frac{1}{n_{\min}}\sum_{(i,j)\in\mathcal{H}} \mathcal{L}_{\text{CE}}\left(\mathbf{S}(:,i,j),\ \mathbf{L}(i,j)\right) \tag{14}$$

where $\mathcal{H}$ is the set of hardest pixels, and the normalization factor $n_{\min} = \max\left(|\mathcal{H}|, n_{\text{threshold}}\right)$ ensures a sufficient number of complex examples are included, with $n_{\text{threshold}} = n_{\text{total}}/16$ and n_{total} being the total number of valid pixels in the image.

TABLE I
HYPERPARAMETER SETTINGS FOR SAM-FUSENET EXPERIMENTS ON CALTECH RGB–T AND PST900

Parameter	Caltech RGB-T	PST900
Image size (pixels)	1024×1024	
Batch size	10	
Training epochs	100	
Loss function	OHEM cross-entropy (no class weights)	
Optimizer	AdamW	
Initial LR	1×10^{-3}	
Weight decay	1×10^{-2}	
LR scheduler	warmup + polynomial decay (power $= 0.9$)	
Warm-up epochs	10	
Warm-up ratio	0.1	
LoRA rank	32	

The overall loss function incorporates the OhemCrossEntropy loss applied to both $\bar{\mathbf{S}}_0$ and $\bar{\mathbf{S}}_1$, as defined in the following equation:

$$\mathcal{L} = w_0\,\mathcal{L}_{\text{Ohem}}\left(\bar{\mathbf{S}}_0,\mathbf{L}\right) + w_1\,\mathcal{L}_{\text{Ohem}}\left(\bar{\mathbf{S}}_1,\mathbf{L}\right) \tag{15}$$

where w_0, $w_1\in\mathbb{R}^+$ are scalar weights that control the relative importance of each loss term.

V. EXPERIMENTS

A. Experiments Settings

1) Datasets: We evaluate the proposed RGB–T semantic segmentation framework on two public benchmarks: the Caltech Aerial RGB–Thermal Dataset in the Wild (Caltech RGB–T) [38] and PST900 [39].

a) Caltech RGB–T: Targets field robotics in natural environments and provides synchronized RGB–T streams together with pixel-wise semantic labels for 10 classes. The labeled set contains 4195 aligned RGB–T image pairs curated from 37 aerial/ground trajectories across rivers, lakes, coasts, deserts, and forests in the United States. The RGB images in the dataset exhibit severe underexposure or extreme contrast conditions. Thermal frames are 16-bit and, per the dataset recommendation, are normalized by rescaling the 1st–99th percentile followed by CLAHE; RGB–T image pairs are stereo-rectified using the released calibrations so that thermal is projected into the RGB frame. We follow these settings and report all numbers on the official splits for reproducibility.

b) PST900: Provides 894 aligned and calibrated RGB–T pairs with per-pixel labels for four subterranean artifact classes, plus 3416 labeled RGB-only images from the same environments. Data were collected with a Stereolabs ZED Mini (RGB) and a FLIR Boson 320 (LWIR). We used the 894 aligned RGB–T pairs for training/evaluation, resize images to a common resolution before batching, and use the class-frequency weighting scheme recommended by the dataset to mitigate foreground–background imbalance.

2) Experiment Settings: We conduct experiments on two RGB–Thermal benchmarks: Caltech RGB–T and PST900, following their official train/test splits. All input images are resized to 1024×1024 pixels. Data preprocessing applies random color jitter, horizontal flip, Gaussian blur, and random

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TABLE II
SEMANTIC SEGMENTATION PERFORMANCE ON THE CALTECH RGB–T DATASET IN TERMS OF MIOU (%) USING RGB–T MODALITIES

Method	Bare Ground	Boulders	Structures	Road	Shrubs	Trees	Sky	Water	Vehicles	Person	mIoU (%)
CWSAM [30]	79.81	87.72	57.04	17.45	55.65	64.05	87.15	94.79	5.61	11.32	44.47
SAM-LoRA [13]	91.84	94.07	70.96	37.53	78.93	85.91	92.45	98.12	25.82	19.14	60.10
CMNeXt [20]	92.29	95.07	65.33	46.21	81.81	89.19	97.51	99.15	37.28	15.07	63.37
MLE-SAM [13]	**94.01**	97.24	82.86	64.68	82.86	91.83	**98.06**	99.31	**64.85**	36.79	70.72
SAM-FuseNet	92.98	**97.80**	**83.86**	**66.79**	**83.79**	**92.02**	97.85	**99.32**	62.90	**42.96**	**74.65**

crop to 1024×1024, followed by channel-wise normalization using dataset mean and standard deviation. Training is performed on an NVIDIA A100 GPU with a batch size of 10. The model is optimized with AdamW (initial learning rate = 1×10^{-3}, weight decay = 1×10^{-2}) for 100 epochs. We adopt the online hard example mining (OHEM) cross-entropy loss without class weights to address class imbalance. A warm-up polynomial learning rate scheduler (power = 0.9) is used, where the learning rate increases linearly during the first 10 epochs with a warm-up ratio of 0.1. The rank of LoRA modules is set to 32. Hyperparameters are summarized in Table I. For evaluation, we report mean intersection-over-union (mIoU).

3) Baseline Implementation Details: We adapt the official code of CMNeXt [20] and CWSAM [30] to Caltech RGB–T and PST900. CMNeXt employs a self-query hub to select informative features from auxiliary modalities and fuses them with the RGB branch, followed by a parallel pooling mixer to extract discriminative cross-modal cues, while RGB remains the reference modality. CWSAM inserts lightweight adapters into the SAM encoder and introduces a class-wise mask decoder that produces dense predictions, transforming the instance-centric SAM architecture into one suitable for semantic segmentation. We further implement SAM-LoRA, which attaches modality-specific LoRA modules to the encoder while freezing the remaining components. In each attention layer, the query and value projections are replaced with low-rank adapters trained independently per modality; the masks predicted by individual adapters are averaged to form a unified representation. Finally, we adopt the MLE-SAM codebase restricted to RGB–T modalities.

B. Comparative Analysis With State-of-the-Art Methods

The quantitative results reported in Table II clearly demonstrate the superiority of the proposed SAM-FuseNet compared with recent SOTA methods. SAM-FuseNet achieves a mIoU of 74.65%, which surpasses the strong baseline MLE-SAM by 3.93% and outperforms CMNeXt by 11.28%. This consistent gain across categories is particularly notable in challenging classes such as Person, where SAM-FuseNet improves IoU from 36.79% with MLE-SAM to 42.96%, and Road, where performance increases from 64.68% to 66.79%.

The superior performance can be attributed to two technical advances. First, the proposed cross-modal fusion mechanism, which integrates RGB–T representations through spatial attention and channel recalibration, enables more reliable feature selection under varying illumination conditions. Second, the hierarchical dual-pathway segmentation head contributes to sharper object boundaries and reduced misclassification by jointly leveraging coarse contextual cues and fine structural details.

TABLE III
SEMANTIC SEGMENTATION PERFORMANCE ON THE PST900 DATASET IN TERMS OF MIOU (%) USING RGB–T MODALITIES

Method	Fire Extinguisher	Backpack	Hand Drill	Rescue Randy	mIoU (%)
MFNet [40]	60.35	64.27	41.13	20.70	57.02
CWSAM [30]	76.27	97.17	68.72	98.99	63.83
LWIR [39]	70.12	69.20	53.60	50.03	68.36
CMNeXt [20]	86.50	99.02	93.24	99.65	88.50
SAM-LoRA [13]	99.15	99.60	99.64	**99.74**	97.11
MLE-SAM [13]	99.27	**99.98**	99.76	99.53	98.58
SAM-FuseNet	**99.65**	99.96	**99.98**	99.54	**98.92**

The class-wise results in Table II show that SAM-FuseNet produces limited improvements over MLE-SAM in categories dominated by broad contextual appearance such as Sky and Water. These regions are spatially homogeneous, cover large portions of the image, and can be reliably separated using low-frequency cues that are already well captured by each modality and by the pretrained backbone. Consequently, the fusion module yields only marginal benefits and may introduce small variations due to cross-branch calibration differences. More substantial gains appear in classes that depend on precise boundary localization and fine-scale structural cues. SAM-FuseNet improves Person from 36.79% to 42.96% and yields moderate increases for Road and Structures, which require the resolution of thin contours, articulated shapes, and small elements that are sensitive to cross-sensor misalignment and partial occlusion. These results reflect the design of the model. Fine-scale fused features enhance spatial precision needed for boundary refinement, whereas coarse and mid-level features mainly encode global semantics and region-level organization that stabilize the overall prediction but contribute less directly to the correction of local boundary errors. Meanwhile, the moderate performance in Vehicles indicates that aerial viewpoints and target size still limit recognition of small, instance-like objects. Addressing these cases likely requires stronger high-resolution supervision and enhanced instance separation strategies while maintaining the advantages provided by multiscale fusion.

Table III presents the semantic segmentation results on the PST900 dataset, where the proposed SAM-FuseNet consistently outperforms existing SOTA methods. SAM-FuseNet achieves a mIoU of 98.92%, surpassing MLE-SAM (98.58%), and SAM-LoRA (97.11%), while substantially outperforming earlier baselines such as CMNeXt (88.50%), LWIR (68.36%),

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TABLE IV

ABLATION STUDY OF MODALITY INPUT, MULTISCALE FEATURES, PIXEL-WISE SPATIAL ATTENTION, AND SEGMENTATION HEAD ON THE CALTECH RGB–T VALIDATION SET. METRIC: MIOU (%)

Modality Input		Multi-scale Feature			Pixel-wise Spatial Attention	Segmentation Head		mIoU
RGB	Thermal	Coarse	Mid	Fine		SAM2 Pathway	Hierarchical Pathway	
✓	-	✓	✓	✓	✓	✓	✓	51.84
-	✓	✓	✓	✓	✓	✓	✓	72.32
✓	✓	✓	-	-	✓	✓	✓	40.86
✓	✓	✓	✓	-	✓	✓	✓	39.40
✓	✓	-	-	✓	✓	✓	✓	69.20
✓	✓	-	✓	✓	✓	✓	✓	72.90
✓	✓	✓	-	✓	✓	✓	✓	69.93
✓	✓	✓	✓	✓	-	✓	✓	73.38
✓	✓	✓	✓	✓	✓	✓	-	70.63
✓	✓	✓	✓	✓	✓	-	✓	51.15
✓	✓	✓	✓	✓	✓	✓	✓	**74.65**

and CWSAM (63.83%). The performance advantage is evident across multiple categories: for Fire Extinguisher, SAM-FuseNet attains 99.65%, improving upon MLE-SAM by 0.38%; for Hand Drill, it reaches 99.98%, the highest among all methods; and for Backpack, it maintains 99.96%, nearly matching the best score of 99.98% from MLE-SAM. These results confirm the robustness of SAM-FuseNet in handling both large rigid objects and fine-structured categories. The gains over MLE-SAM, though numerically smaller than on Caltech RGB–T, highlight the efficiency of the proposed design in extracting complementary modality cues, even in scenarios where baseline methods already deliver strong results.

It is worth noting that the margin of improvement is limited in categories such as Backpack and Rescue Randy, where both MLE-SAM and SAM-FuseNet achieve near-perfect segmentation. This suggests that PST900's relatively clean modality alignment and large-scale targets reduce the difficulty of the task. Nevertheless, the consistent improvements across categories, combined with the best overall mIoU, validate that SAM-FuseNet's generalization ability across diverse datasets.

C. Ablation Study

The ablation results in Table IV provide a detailed quantitative evaluation of each component within the proposed SAM-FuseNet framework. When trained with the unimodal configurations using only RGB or only thermal input yield mIoU scores of 51.84% and 72.32%, respectively, highlighting the dominant contribution of the thermal modality under low-light or high-contrast conditions.

Regarding multiscale feature integration, using only coarse features yields an mIoU of 40.86%, showing that low-resolution semantic context alone cannot reliably localize boundaries in RGB–T scenes. Extending this setting with midlevel features reduces performance to 39.40%, which indicates that midscale responses, when operating without fine-grained cues, introduce additional intraclass variation and modality-specific inconsistencies that coarse semantics cannot resolve. In contrast, using only fine features achieves 69.20% mIoU, demonstrating the importance of high-resolution spatial detail for boundary localization, although this configuration still lacks stable global context.

The two-scale results reveal the distinct functional contributions of different feature levels. Combining mid and fine features attains 72.90%, the strongest performance among all two-scale variants. This improvement shows that mid-level representations are most effective when paired with fine-scale activations: the mid-scale features encode object-part structure and region-level geometry that regularize the high-frequency noise present in fine features, while the fine-scale responses sharpen boundaries and correct local artifacts. By comparison, combining coarse and fine features yields 69.93%, only marginally higher than the fine-only setting. The gap relative to the mid and fine configuration suggests that very low-resolution coarse maps, although semantically informative, are less capable than mid-level features of preserving spatial organization and part-level consistency, and may introduce excessive smoothing that degrades edge precision.

These observations support the inclusion of both coarse and mid-scale features in the framework. Coarse features provide broad semantic context that stabilizes predictions under significant illumination variation and cross-modal inconsistencies, including regions affected by shadows or thermal saturation. Mid-level features serve as an intermediate representation that captures object-part structure and local shape cues over a moderate receptive field, which is critical for aligning RGB–T signals at a scale where geometry is still preserved. This alignment, in turn, guides fine-scale filters to focus on true physical boundaries rather than modality-specific noise. When all three scales are jointly integrated, the model reaches 74.65% mIoU, confirming that a hierarchical multiscale feature design is essential for robust RGB–T fusion.

The dual-pathway segmentation head reveals further insights. The first pathway alone achieves 70.63%, while the second pathway alone produces 51.15%. Although the second pathway is weaker in isolation due to its reliance on hierarchical fusion, its complementary role becomes evident when integrated with the first pathway, leading to superior overall performance.

Fig. 4 illustrates the hierarchical feature representations extracted from RGB, Thermal, and fused modalities at coarse, middle, and fine scales. The first two rows show that single-modality features often exhibit incomplete or noisy structural information. For example, RGB features are sensitive to illumination conditions and contain blurred textures, while Thermal features capture scene geometry but suffer from reduced detail in fine regions. In contrast, the fused features in the third row demonstrate complementary integration, where object boundaries become sharper and semantic structures are more consistent across scales. Notably, the fused feature retains global contours from Thermal while preserving fine-grained details from RGB, leading to semantically coherent and spatially precise representations. This evidence highlights the effectiveness of the proposed fusion mechanism in mitigating modality-specific limitations and ensuring robustness to visual degradations, thereby supporting more reliable downstream segmentation compared with single-modality or naïve fusion baselines.

As illustrated in Fig. 5, the proposed dual-pathway design yields superior segmentation quality compared with

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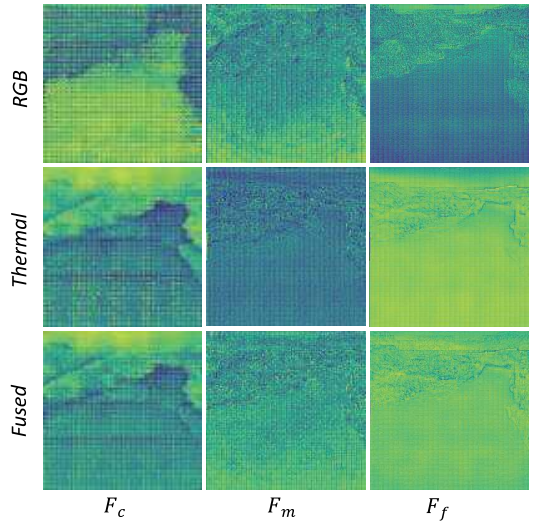


Fig. 4. Feature representations extracted from RGB, Thermal, and fused modalities at multiscales, where each tensor is reduced to a 2-D heatmap by channel averaging.

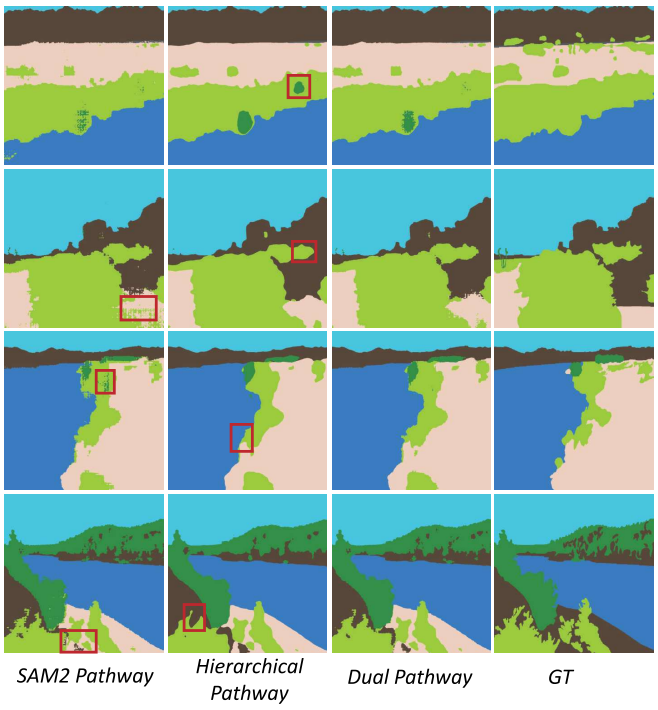


Fig. 5. Qualitative comparison of segmentation results across the first pathway, second pathway, and the proposed dual-pathway design, with ground truth provided for reference.

single-pathway variants. The first pathway produces coarse predictions with noticeable boundary artifacts and class confusion, particularly in regions containing shrubs and small structures. The second pathway improves spatial coherence but still suffers from local misclassifications, as highlighted by the red bounding boxes. By contrast, the dual-pathway strategy effectively integrates complementary predictions, producing sharper object boundaries and reducing spurious noise in homogeneous regions such as sky and water. The consistency of the results across diverse scenes demonstrates the robustness of the dual-pathway formulation, validating its ability to capture both global context and fine-scale details. These visual results substantiate the quantitative gains reported in the main experiments and confirm that the dual-pathway head plays a crucial role in improving segmentation accuracy and generalization.

TABLE V

MISSING-MODALITY EVALUATION FOR MODELS TRAINED WITH PAIRED RGB–T INPUTS. AT TEST TIME, RGB ZEROS THE THERMAL MODALITY, THERMAL ZEROS THE RGB MODALITY, AND RGB–T USES BOTH MODALITIES. AVERAGE IS THE MEAN OF THE THREE SETTINGS

Method	RGB	Thermal	RGB-T	Average
CWSAM [30]	11.92	28.65	44.47	28.35
SAM-LoRA [13]	**15.17**	49.06	60.10	41.44
CMNeXt [20]	8.23	26.80	63.37	32.80
MLE-SAM [13]	9.84	67.15	70.72	49.24
SAM-FuseNet	10.96	**70.86**	**74.65**	**52.16**

D. Qualitative Results

Fig. 6 presents the qualitative comparison of semantic segmentation results produced by the proposed SAM-FuseNet against several SOTA approaches. The first two rows illustrate cases where the RGB modality is well-exposed, while the last two rows correspond to scenarios where the RGB modality suffers from underexposure. Red bounding boxes highlight failure cases of competing methods. Under normal conditions, SAM-FuseNet generates sharper object boundaries and significantly reduces misclassification compared to CWSAM, SAM-LoRA, CMNeXt, and MLE-SAM. Competing models often mis-segment small structures or merge object boundaries into background regions, whereas SAM-FuseNet, benefiting from the dual-pathway segmentation head, yields accurate delineation of both large-scale regions and fine structures. More importantly, under underexposed conditions where the RGB modality becomes unreliable, SAM-FuseNet maintains robust predictions by effectively leveraging complementary Thermal information through its spatial attention fusion and channel recalibration mechanisms. While other methods show pronounced degradation, producing fragmented or distorted masks, SAM-FuseNet demonstrates consistent robustness with accurate scene layouts and reduced noise. These results highlight the model's strong resilience to visual degradations and its ability to exploit RGB–T complementarity for reliable semantic segmentation across diverse conditions.

To evaluate the robustness of SAM-FuseNet under modality dropout and to compare it with competing architectures, we conduct a missing-modality study on the Caltech RGB–T validation set, as reported in Table V. Each model is tested with structurally valid inputs in which the absent modality is replaced by a zero-valued tensor that matches the original spatial and channel dimensions. This procedure retains the network's computational graph and fusion structure while ensuring that the missing stream contributes no informative

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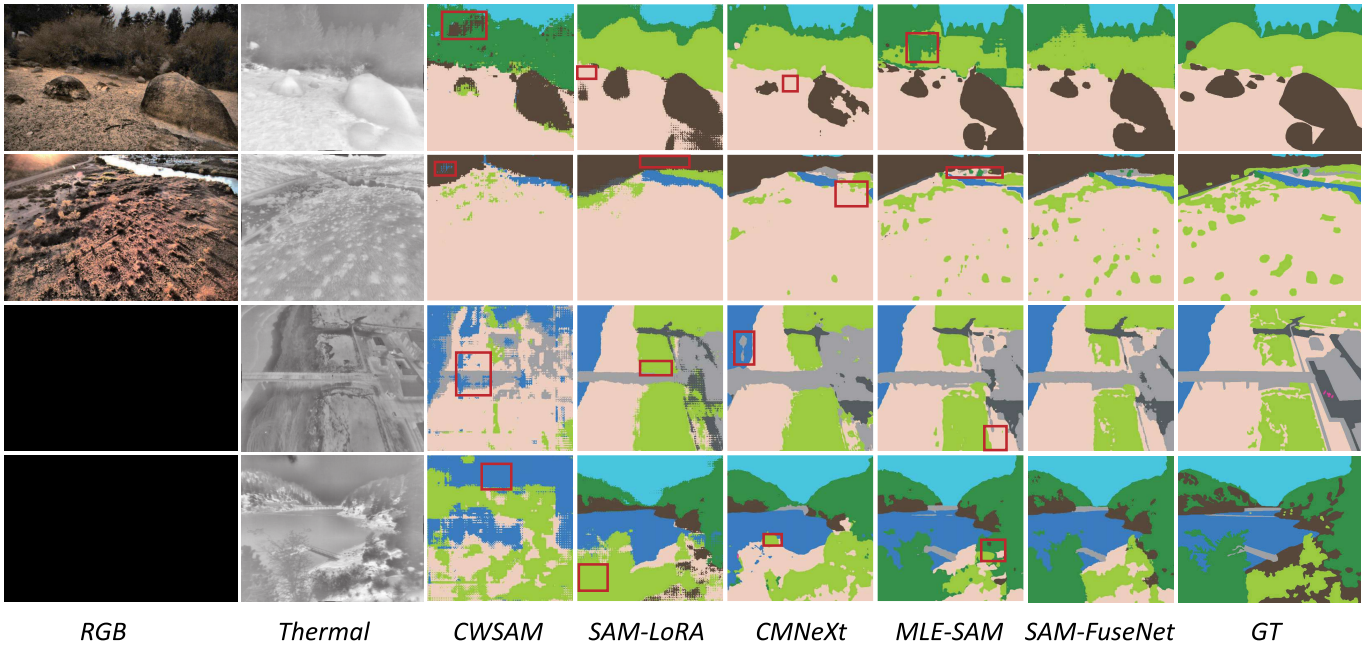


Fig. 6. Qualitative comparison of segmentation results under normal and underexposed RGB conditions.

content. Across all methods, removing a modality leads to a notable reduction in mIoU, yet the magnitude and structure of this reduction vary considerably among architectures, revealing differences in how they encode and fuse signals.

CWSAM drops from 44.47% to 11.92% and 28.65% when evaluated with RGB-only and Thermal-only inputs. SAM-LoRA obtains the strongest RGB-only score among the baselines with 15.17%, achieves 49.06% with Thermal-only, and reaches 60.10% with RGB–T input, although it lacks explicit cross-modal modeling and thus falls behind more specialized fusion approaches. CMNeXt performs well when both modalities are available with 63.37%, but shows severe degradation with single-modality input, with 8.23% for RGB-only and 26.80% for Thermal-only. This decline indicates that its convolutional fusion is strongly tied to paired RGB–T structure and becomes unstable when one branch is absent. MLE-SAM and SAM-FuseNet, both incorporating SAM2 with modality-specific adaptation, show stronger resilience. MLE-SAM reaches 67.15% with Thermal-only and 70.72% with RGB–T input, with its RGB-only score remains low at 9.84%, suggesting a strong bias toward the thermal stream. SAM-FuseNet improves the Thermal-only score to 70.86% and sets a new best fused performance of 74.65%, while also maintaining a comparable RGB-only score of 10.96%. Its balanced behavior yields the highest overall average of 52.16%.

The asymmetric degradation between RGB-only and Thermal-only settings is largely driven by the characteristics of the aerial RGB–T data. A substantial number of RGB frames are severely underexposed, causing most pixel values to cluster near zero. Under such conditions, the patch embeddings derived from the RGB encoder collapse to low-variance tokens with weak structural information. After normalization, these embeddings appear nearly uniform, limiting the transformer's ability to form meaningful spatial relations and leaving the decoder with little geometric or textural evidence for segmentation. By contrast, thermal imagery preserves reliable contrast across a wide range of illumination conditions, providing stable gradients, clear boundaries, and consistent object signatures. As a result, thermal-only inference retains significantly higher accuracy for models with strong thermal adaptation, such as MLE-SAM and SAM-FuseNet.

SAM-FuseNet demonstrates improved robustness because its architecture explicitly accounts for heterogeneous sensing quality. The LoRA-enhanced modality-specific encoders allow each branch to specialize to its own signal statistics while preserving the strong priors of SAM2. The pixel-wise spatial attention module, combined with SE-based channel recalibration, learns to upweight thermal features when RGB is unreliable and avoids propagating noise from underexposed regions. During training, this mechanism consistently identifies thermal features as more informative in low-light scenes, enabling effective cross-modal alignment. The dual-pathway segmentation head further stabilizes output under missing-modality conditions by aggregating multi-scale context and refining boundaries using SAM2-guided cues. Together, these components enable SAM-FuseNet to produce high-quality predictions in the full RGB–T setting and to remain comparatively stable even when one modality is removed, leading to the best fused and average performance among all evaluated methods.

E. Efficiency Analysis for Aerial Robotics

Table VI summarizes the throughput, computational cost, and trainable parameter count (in millions) for several models at input resolutions of 1024×1024 and 512×512. At 1024×1024, CWSAM attains 7.89 frames/s with 12.64 GFLOPs and 4.20 M parameters, while SAM-LoRA achieves a higher 10.93 frames/s at nearly the same computational cost but with only 2.85 M parameters, demonstrating better parameter efficiency. CMNeXt delivers the highest

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TABLE VI
THROUGHPUT AND COMPUTATIONAL COST AT DIFFERENT INPUT RESOLUTIONS

Method	Resolution	FPS ↑	GFLOPs ↓	Params (M) ↓
CWSAM [30]	1024 × 1024	7.89	12.64	4.20
SAM-LoRA [13]	1024 × 1024	10.93	12.40	**2.85**
CMNeXt [20]	1024 × 1024	**28.31**	63.24	10.29
MLE-SAM [13]	1024 × 1024	3.87	65.90	10.40
SAM-FuseNet	1024 × 1024	8.37	38.26	4.71
SAM-FuseNet	512 × 512	23.15	**9.93**	4.71

throughput of 28.31 frames/s but incurs a heavy computational burden of 63.24 GFLOPs and 10.29 M parameters, making it less suitable for resource-limited scenarios. In contrast, MLE-SAM records the lowest efficiency, reaching only 3.87 frames/s at 65.90 GFLOPs and 10.40 M parameters. The proposed SAM-FuseNet achieves 8.37 frames/s with 38.26 GFLOPs and 4.71 M parameters at 1024 × 1024, representing a balanced tradeoff as it surpasses CWSAM and MLE-SAM in computational efficiency while maintaining a much lower computational cost than CMNeXt. When the input resolution is reduced to 512×512, SAM-FuseNet achieves 23.15 frames/s with only 9.93 GFLOPs while maintaining the same parameter count. This result demonstrates a favorable balance between throughput, computational complexity, and model size. Among all methods, SAM-FuseNet at 512 × 512 provides the most practical compromise, attaining near real-time performance with manageable computational requirements.

For aerial robotics, onboard perception systems must satisfy stringent real-time constraints due to limited computational and power budgets. Prior studies indicate that UAV-based segmentation systems typically require 20–30 frames/s to support timely perception and control on embedded devices [41]. Under this criterion, SAM-FuseNet's 23.15 frames/s at 512 × 512 satisfies the lower bound of real-time performance, validating its suitability for onboard deployment. Although the 1024 × 1024 configuration achieves only 8.37 frames/s, it provides higher fidelity when computational resources allow. In UAV operations characterized by motion blur, rapid flight dynamics, and visually complex environments containing rooftops, vehicles, and pedestrians, the 512 × 512 resolution offers a practical balance between spatial detail and real-time capability. In comparison, CMNeXt's high throughput at 1024 × 1024 comes with excessive computational cost of 63.24 GFLOPs, and SAM-LoRA, though lightweight, lacks sufficient speed for high-velocity UAV navigation. Consequently, SAM-FuseNet achieves the most favorable compromise among speed, computational load, and parameter efficiency, making it well-suited for real-time aerial robotics applications.

VI. CONCLUSION AND FUTURE WORK

In this work, we introduced SAM-FuseNet, a novel framework for RGB–T semantic segmentation that integrates modality-specific SAM2-based encoders with LoRA, a spatial attention fusion mechanism with channel recalibration, and a dual-pathway segmentation head. Methodologically, the proposed design enables efficient modality adaptation, adaptive feature alignment, and joint exploitation of coarse contextual cues and fine structural details. Empirically, SAM-FuseNet consistently improves segmentation accuracy over SOTA baselines, achieving gains of up to 3.93% in mIoU while requiring less than half the trained parameters of competing approaches. Qualitative and quantitative results further demonstrate sharper object boundaries, robustness to illumination degradations, and resilience to modality imbalance.

The broader significance of SAM-FuseNet lies in its ability to address two persistent challenges in multimodal perception: efficient adaptation of large pretrained models and robust cross-modal fusion under varying conditions. By balancing accuracy with efficiency, the framework has strong potential for deployment in real-world robotic applications where computational resources and environmental variability pose critical constraints.

Future extensions of SAM-FuseNet include incorporating additional sensing channels such as depth, event, LiDAR, or hyperspectral inputs to broaden applicability in heterogeneous aerial environments. The missing-modality analysis also reveals that robustness can be further improved by training with modality dropout and exposure perturbation, enabling the fusion module to handle unreliable or absent inputs more effectively. Incorporating lightweight exposure-correction or quality-estimation modules may strengthen RGB-only performance under severe underexposure. Finally, designing the framework to operate reliably with arbitrary subsets of three or more modalities would enhance its practicality in real-world deployments where sensor failures occur independently.

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